

MONIL ARORA

Vancouver, BC

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Education

University of British Columbia, Vancouver

Expected 2027

Bachelor of Applied Science in Electrical Engineering

Relevant Coursework: Machine Learning with Engineering Applications, Digital Systems Design, Computing Systems I, Signals and Systems, Introduction to VLSI Systems

Work Experience

Laser Zentrum Hannover (LZH)

Jan – Aug 2026

Research Intern

Hannover, Germany

- Developed **STM32** flight firmware in **C++/FreeRTOS** across two hardware targets (F401CC dev board, F103C8 mission PCB), running **10 concurrent tasks** that encode **1 Hz** telemetry as **CCSDS Space Packets** over **115.2 kbaud RS-422**.
- Fit the complete CCSDS stack, laser PWM, and FreeRTOS kernel inside the F103's tight **20 KB RAM** budget, clearing a **784-byte** overflow by re-budgeting heap and per-task stacks to ship with **240 B** to spare.
- Replaced a proprietary byte protocol with a standards-compliant **COP-1 FARM-1** receiver publishing **CLCW** in the TM frame OCF, gaining uplink retransmit and de-duplication at a **1.4%** bandwidth cost.
- Integrated **LittleFS** non-volatile storage for boot count and RTC time, scoping writes to boot-time only after determining that per-sample logging would exhaust flash endurance in **20 days**.
- Implemented end-to-end **image download**, streaming a 7 KB JPEG as **91 79-byte APID 131** chunks from an 8 KB SRAM buffer at **1.5 KB/s** into YAMCS object storage via a Python **WebSocket** reassembler.
- Built custom **Java data links** for **YAMCS** (SLIP-over-UDP TM, **COP-1 FOP TC** uplink) and an **XTCE database** decoding **APID 130 TM / 100 TC** with threshold alarms and **command verifiers**.

Sharang Shakti Pvt Ltd

May – Jun 2024

Embedded Signal Processing Engineer

Gurugram, India

- Brought up an **AD9910 DDS** evaluation board for the **digitization of a legacy analog radar system**, synthesizing high-frequency transmit waveforms via profile registers, **RAM playback**, and the **digital ramp generator**; wrote Python to build **14-bit DAC sample tables** and stabilized the reference-clock **PLL** to sharpen resolution and suppress **phase noise**.

Projects

45 nm Dot-Product Accelerator | *SystemVerilog, Cadence Genus, Innovus, ModelSim*

Sep – Dec 2025

- Designed a four-element **MAC dot-product accelerator** in **SystemVerilog** and took it **RTL-to-GDSII** on a **45 nm** PDK with **Genus** synthesis, **Innovus** place-and-route, and **clock tree synthesis** (**104 ps** skew, 1,320 flops).
- Characterized power from measured switching activity, running **gate-level simulation** with back-annotated **SDF** delays in **ModelSim** and converting **VCD to SAIF**, which corrected a 1.36 mW estimate to **68.7 μ W**.

Self-Balancing Robot | *C++, Arduino, ESP32, IMU, PID, Embedded Systems*

Jan – Apr 2025

- Prototyped a two-wheeled **self-balancing robot** on an **Arduino Nano 33 BLE** with a **BMI270 IMU** and **motor drivers**, closing a **real-time PID loop** with **complementary-filter** sensor fusion and **PWM** actuation for balance up to **45°** of tilt; verified by **oscilloscope analysis**, achieving approximately **30% improvement in stability**.
- Extended the platform with an **ESP32 BLE** remote transmitting drive commands and displaying live RPM, tilt angle, and sonar distance on an **LCD**.

Deep Neural Network Accelerator | *Verilog, C, Nios II, FPGA*

Oct – Nov 2024

- Accelerated MNIST inference on a **DE1-SoC** FPGA running an embedded **Nios II** system, with **Q16.16 fixed-point arithmetic** and **ReLU** activation.
- Architected a **system-on-chip (SoC)** around **Avalon** dot-product units and an **SDRAM interface**, applying **Quartus timing constraints** to hold stable operation at **50 MHz**.

Reflow Oven Controller | *Assembly, 8051*

Jan – Feb 2024

- Programmed a **microcontroller-based reflow oven controller** on the Nuvoton **N76E003 (8051 core)** entirely in **assembly language**, sequencing a six-state **soldering temperature profile** from **25°C to 240°C** to within $\pm 3^\circ\text{C}$, with **fail-safe shutdowns and alerts** and a serial **PC monitoring interface**.

Technical Skills

Languages: C/C++, Python, Java, Assembly, Verilog/SystemVerilog

Tools: YAMCS, FreeRTOS, PlatformIO, Maven, Quartus, ModelSim, CrossIDE, Cadence, MATLAB, LTspice, Git

Hardware: STM32, ESP32, EFM8, Arduino, DE1-SoC, Raspberry Pi, IMU Sensors, Motor Drivers

ML/Data: PyTorch, scikit-learn, NumPy, Matplotlib, CVXPY